Project Design Phase-II

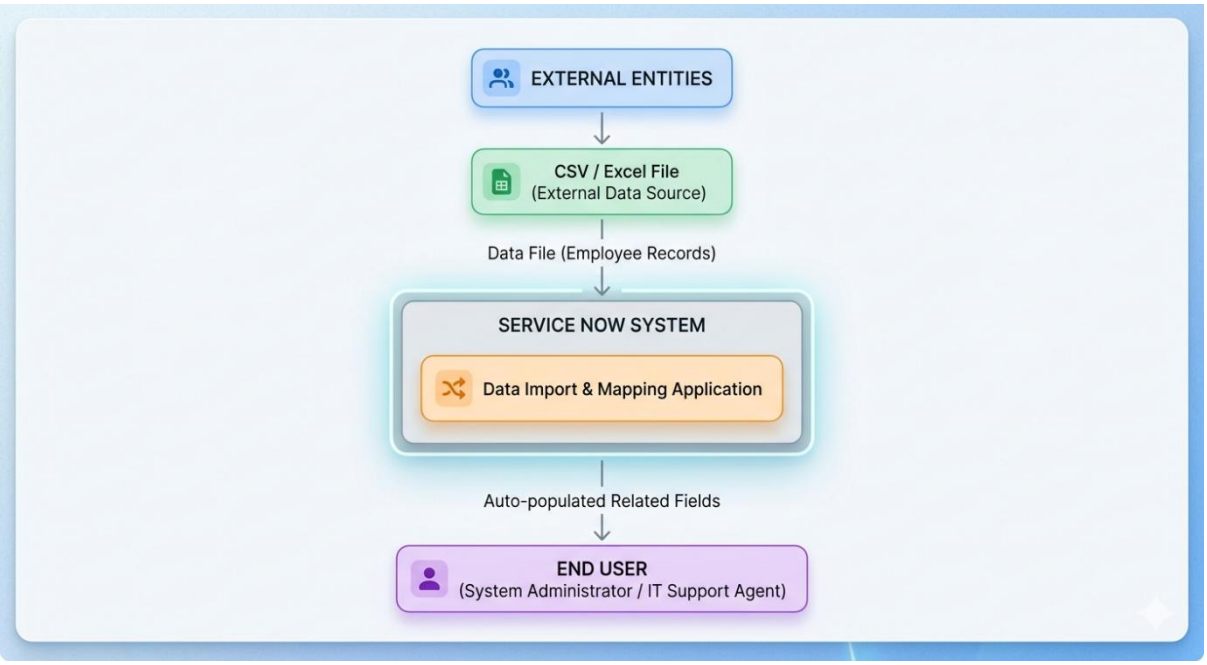
Data Flow Diagram & User Stories

Date	05 April 2026
Team ID	SWTID-2026-5577
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

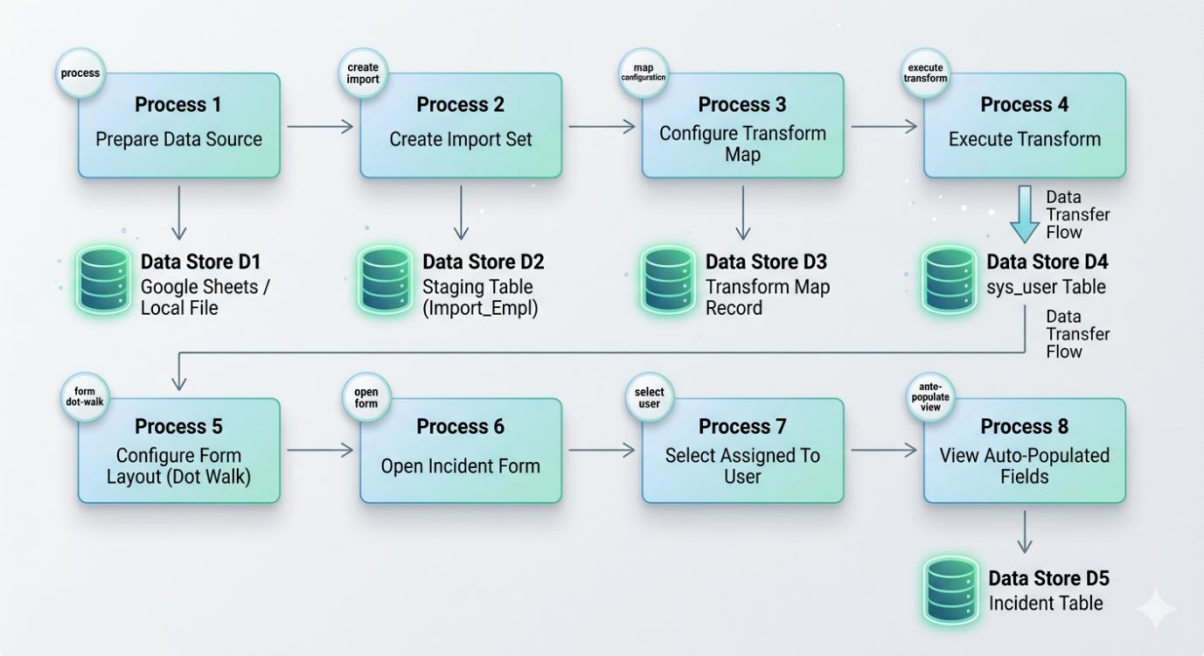
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Level 0 DFD (Context Diagram)



Level 1 DFD (Detailed Process Flow)



Detailed Data Flow Description

Step	Process	Input	Output	Data Store
1	User prepares data in Google Sheets/Excel	Manual data entry	Structured CSV/Excel file	**D1:** Local File System
2	User uploads file to ServiceNow Import Set	CSV/Excel file	Staging table records	**D2:** Import Employee Table
3	System auto-maps fields using Transform Map	Staging table fields	Field mapping configuration	**D3:** Transform Map Record
4	System executes transformation	Mapped field data	Inserted/Updated records in target table	**D4:** sys_user Table

Step	Process	Input	Output	Data Store
5	Admin configures Incident form with dot-walked fields	Form configuration settings	Updated form layout	**N/A**
6	Agent opens new Incident form	User action	Blank incident form	**N/A**
7	Agent selects "Assigned To" user	User selection (sys_id)	Trigger dot-walking lookup	**D4:** sys_user Table
8	System auto-populates related fields	User reference	Department, Email, and Manager values	**N/A**

User Stories

Following are the user stories for the proposed solution based on different user roles.

User Role 1: System Administrator

Story ID	User Story	Acceptance Criteria	Priority
US-ADMIN-1	As a System Administrator, I want to upload employee data from CSV/Excel files so that I can bulk import records without manual entry.	Administrator can select and upload a local CSV/Excel file into the ServiceNow instance.	High
US-ADMIN-2	As a System Administrator, I want the system to automatically create a staging table when I upload data so	A staging table (e.g., Import Employee) is generated with records visible for verification.	High

Story ID	User Story	Acceptance Criteria	Priority
	that I can review records before final import.		
US-ADMIN-3	As a System Administrator, I want to use Auto Map Matching to map source fields to target fields so that I save time on manual field mapping.	Fields with identical names in the source and target (*sys_user*) are mapped automatically.	Medium
US-ADMIN-4	As a System Administrator, I want to manually map unmapped fields (like Manager) using Mapping Assist so that all relationships are correctly established.	Reference fields and complex mappings are successfully configured and saved.	High
US-ADMIN-5	As a System Administrator, I want to execute the Transform Map and see transformation status so that I know if data was imported successfully.	The transformation completes and displays a "Success" or "Completed" status.	High
US-ADMIN-6	As a System Administrator, I want to configure form layouts by adding dot-walked fields so that end users can see related data automatically.	Related data (e.g., Manager's Email) appears on the Incident form when a user is selected.	High
US-ADMIN-7	As a System Administrator, I want error logs when transformation fails so that I can debug and fix data issues.	The Import Set Run record provides detailed error descriptions for any failed rows.	Medium

User Role 2: IT Support Agent

Story ID	User Story	Acceptance Criteria	Priority
US-AGENT-1	As an IT Support Agent, I want to create a new incident ticket so that I can track user issues.	Agent can open the Incident form and save a new record.	High
US-AGENT-2	As an IT Support Agent, I want the Department field to auto-populate when I select an "Assigned To" user so that I don't have to look it up manually.	The Department field updates instantly upon user selection via dot-walking.	High
US-AGENT-3	As an IT Support Agent, I want the Email field to auto-populate when I select a user so that I can quickly contact them.	The Email field accurately displays the value from the *sys_user* record.	Medium
US-AGENT-4	As an IT Support Agent, I want the Manager field to auto-populate when I select a user so that I know who to escalate issues to.	The Manager's name is displayed as a reference link on the Incident form.	High
US-AGENT-5	As an IT Support Agent, I want all related user information to be accurate and up-to-date so that I can make informed decisions.	Data displayed on the form matches the most recent import from the source file.	High

User Role 3: IT Manager

Story ID	User Story	Acceptance Criteria	Priority
US-MGR-1	As an IT Manager, I want all employee data to be consistent across the system so that reports are accurate.	Data in the *sys_user* table matches the source CSV/Excel file with zero discrepancies.	High
US-MGR-2	As an IT Manager, I want the data import process to be automated so that my team spends less time on manual data entry.	The system supports recurring Data Source imports without requiring manual field mapping each time.	Medium
US-MGR-3	As an IT Manager, I want relationship mapping between users, departments, and managers to be maintained automatically so that organizational hierarchy is always correct.	Reference fields (Manager, Department) correctly link to their respective tables during transformation.	High
US-MGR-4	As an IT Manager, I want to audit import logs so that I can track who imported what data and when.	Import Set Run records and transformation logs are accessible and show the "Created By" and "Created" timestamps.	Medium

User Role 4: HR Administrator

Story ID	User Story	Acceptance Criteria	Priority
US-HR-1	As an HR Administrator, I want to export employee data from our HR system into CSV format so that it can be imported into ServiceNow.	Data is exported in a structured CSV/Excel format with defined column headers (Name, Email, Dept).	High
US-HR-2	As an HR Administrator, I want updated employee records to be reflected in ServiceNow automatically so that IT always has current information.	System triggers a data load and transformation from the provided source to the *sys_user* table.	Medium
US-HR-3	As an HR Administrator, I want to validate that all imported employee records have correct email formats and manager assignments so that data quality is maintained.	Staging table records are reviewed to ensure consistent naming and valid email formats.	High

Acceptance Criteria for Key User Stories

Story ID	Acceptance Criteria
US-ADMIN-1	✓ CSV/Excel file upload completes without error ✓ All records appear in staging table ✓ File format validation works
US-ADMIN-4	✓ Manager field maps to reference field type ✓ Relationship is stored as sys_id ✓ Dot walking can traverse Manager relationship
US-AGENT-2	✓ Selecting user automatically fills Department ✓ Department value matches imported data ✓ Field is read-only (auto-populated)
US-AGENT-4	✓ Selecting user automatically fills Manager name ✓ Manager name is clickable (reference link) ✓ Manager record can be opened from incident